Dokumentowe bazy danych – MongoDB

Ćwiczenie 1

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Test połączenia

- informacja o wersji

```
db.version()
```

- wynik

```
{
  "result": "7.0.7"
}
```

- lista baz danych na serwerze

```
show dbs;
show databases;
db.adminCommand('listDatabases');
```

- wynik

```
[
  {
    "empty": false,
    "name": "admin",
    "sizeOnDisk": 40960
  },
  {
    "empty": false,
    "name": "config",
    "sizeOnDisk": 98304
  },
  {
    "empty": false,
```

```
    "name": "local",
    "sizeOnDisk": 40960
  }
]
```

- wybór bazy danych
  - baza o nazwie univ

```
use univ;
```

- informacja o bieżącej bazie danych

```
db;
```

- informacja o kolekcjach

```
show collections;
db.getCollectionNames();
```

Proste operacje/zapytania

- MongoDB simple query
- https://www.mongodb.com/docs/manual/tutorial/query-documents/

Przykład 1

proste operacje na dokumentach

- wstaw/stwórz pierwszy dokument
  - w tym momencie tworzona jest baza danych i kolekcja (jeśli wcześniej nie istniały)

```
db.student.insertOne(
    {
        "student_id": 1,
        "firstname": "John",
        "lastname": "Gold",
        "age": 25
```

```
    }
);
```

- sprawdź listę baz i kolekcji

```
show databases;
```

- wynik

```
[
  {
    "empty": false,
    "name": "admin",
    "sizeOnDisk": 40960
  },
  {
    "empty": false,
    "name": "config",
    "sizeOnDisk": 98304
  },
  {
    "empty": false,
    "name": "local",
    "sizeOnDisk": 40960
  },
  {
    "empty": false,
    "name": "univ",
    "sizeOnDisk": 8192
  }
]
```

```
show collections;
```

- wynik

```
[
  {
    "badge": "",
    "name": "student"
  }
]
```

- wyszukaj dokumenty w kolekcji `employees`

```
db.student.find();
db.student.find({});
```

- wynik

```
[
  {
    "_id": {"$oid": "67e86e45b1c69e3d7e42c9c0"},
    "age": 25,
    "firstname": "John",
    "lastname": "Gold",
    "student_id": 1
  }
]
```

- stwórz indeks zapewniający unikalność atrybutu `"student_id"`

```
// tworzenie indeksu
db.student.createIndex({ "student_id" : 1 }, { "unique": true });

// inf. o indeksach w kolekcji student
db.student.getIndexes();

// usunięcie indeksu o nazwie
db.student.dropIndexes("student_id_1");
```

- wstaw kilka dokumentów do kolekcji `student`
    - np

```
db.student.insertOne(
    {
        "student_id": 2,
        "firstname": "James",
        "lastname": "Bond"
    }
);


db.student.insertOne(
    {
        "firstname": "John",
        "lastname": "Bond"
```

```
    }
);
```

- wyszukaj dokumenty w kolekcji `student`

```
db.student.find();
db.student.find({});
db.student.find({"firstname": "John"});
db.student.find({"student_id": 1});

db.student.find({"_id": ObjectId("67e669647013d10c86e71c87")});
```

- zmodyfikuj wybrane dokumenty
    - po wykonaniu każdego z przykładów sprawdź wynik za pomocą `db.student.find()`
- np

```
db.student.updateOne(
    {"student_id" : 1},
    {
        $set: {
            "firstname": "Adam",
            "lastname": "Silver"
        }
    }
)
```

```
db.student.updateOne(
    {"student_id" : 1},
    {
        $inc: {"age": 5}
    }
)
```

```
db.student.updateOne(
    {"student_id" : 1},
    {
        $set: {
            "firstname": "John",
            "lastname": "Gold"
        },
        $inc: {
            "age": -2
```

```
        }
    }
)
```

```
db.student.updateOne(
    {"student_id" : 2},
    {
        $set: {
            "age":  19
        }
    }
)
```

```
db.student.updateOne(
    {"student_id" : 2},
    {
        $unset: {
            "age": ""
        }
    }
)
```

- usuń wybrane dokumenty
    - po wykonaniu każdego z przykładów sprawdź wynik za pomocą `db.student.find()`
- np

```
db.student.deleteOne({"student_id": 1})
db.student.deleteMany({"student_id": 1})
db.student.deleteOne({"_id": ObjectId("67e669647013d10c86e71c87")});
```

- `replace`

```
db.student.find({"student_id": 1});
```

```
db.student.replaceOne(
    {
        "student_id": 1
    },
    {
        "student_id": 1,
```

```
        "firstname": "Jan",
        "lastname": "Kowalski",
        "hobby": "Ski"
    }
);
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "hobby": "Ski",
    "lastname": "Kowalski",
    "lirstlame": "Jan",
    "student_id": 1
  }
]
```

- tablice
    - `grades` - tablica ocen studenta

```
db.student.updateOne(
    {
        "student_id": 1
    },
    {
        $set : {
            "grades" : [3, 4, 5, 6, 8]
        }
    }
);
```

```
db.student.find({"student_id": 1});
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [3, 4, 5, 6, 8],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
```

```
  }
]
```

- średnia ocena

```
db.student.aggregate([
    {
        $match : {
            "student_id": 1
        }
    },
    {
        $project: {
            "firstname": 1,
            "lastname": 1,
            "student_id": 1,
            "grades": 1,
            "averagegrade": { $avg: "$grades" }
        }
    }
])
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "averagegrade": 5.2,
    "firstname": "Jan",
    "grades": [3, 4, 5, 6, 8],
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

- dodanie elementu do tablicy

```
db.student.updateOne(
    { student_id: 1 },
    { $push: { grades: 10 } }
);
```

```
db.student.find({"student_id": 1});
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [3, 4, 5, 6, 8, 10],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

- modyfikacja elementu w tablicy
  - indeksy zaczynają się od 0

```
db.student.updateOne(
    { student_id: 1 },
    { $set: { "grades.2": 12 } }
)
```

```
db.student.find({"student_id": 1});
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [3, 4, 12, 6, 8, 10],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

- `$map` - wykonanie operacji dla każdego elementu tablicy
  - https://www.mongodb.com/docs/manual/reference/operator/aggregation/map/

- w przykładzie poniżej
    - dzielimy każdą ocenę przez 20

```
db.student.updateOne(
    { student_id: 1 },
    [
        {
            $set: {
                grades: {
                    $map: {
                        input: "$grades",
                        as: "g",
                        in: { $divide: ["$$g", 20] }
                    }
                }
            }
        }
  ]
)
```

```
db.student.find({"student_id": 1});
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
    "grades": [0.15, 0.2, 0.6, 0.3, 0.4, 0.5],
    "hobby": "Ski",
    "lastname": "Kowalski",
    "student_id": 1
  }
]
```

Zadanie 1

- Wykonaj kilka własnych eksperymentów z operacjami CRUD
- przydatne linki
    - https://www.mongodb.com/docs/manual/crud/
    - https://www.mongodb.com/docs/manual/reference/method/db.collection.find/

- https://www.mongodb.com/docs/manual/reference/method/db.collection.insertOne/
- https://www.mongodb.com/docs/manual/reference/method/db.collection.updateOne/
- https://www.mongodb.com/docs/manual/reference/method/db.collection.deleteOne/

Zadanie 1 - rozwiązanie - Michał Kościanek, code review - Michał Mąka

> Wyniki:
>
> przykłady, kod, zrzuty ekranów, komentarz ...

```
// FIND
db.customers.find({Country:"UK"})

//kolumny w tabeli "rozciągają" się, tworząc wielki zbiór kolumn, a rekordy
// nie mające tej kolumny maja tam null
db.customers.find({Region:null})


//INSERT

//można dodać pusty rekord
db.customers.insertOne({})
db.customers.insertMany([{},{},{}])

//poprawna wersja użycia insertOne, kolejność atrybutów w dodawaniu
// nie ma znaczenia
db.customers.insertOne({
  Address:"Obere Str. 57",
  City:"Amsterdam",
  CompanyName:"NHL",
  ContactName:"Johan Cruyff",
  Country:"Netherland",
  CustomerID:"NHL",
  Phone:"3434-23425",
  PostalCode:"43-531"})


db.customers.find()

//REPLACE i UPDATE

//przy zamianie rekordów trzeba pamietać, że nie wymieniamy tylko atrybutu,
// a całą kolekcje,bo możemy dostać rekord z wypełnionym tylko jednym atrybutem
db.customers.replaceOne({City:"Bern"},{City:"Locarno"})

//chcąc zmienić tylko jeden atrybut w wierszu trzeba zrobić to
db.customers.updateOne({City:"Madrid"},{$set:{City:"Barcelona"}})

//DELETE
//usunięcie wszystkich rekordów
```

```
db.customers.deleteMany({})

//poprawne usunięcie rekordu w bazie danych
db.customers.deleteOne({City:"Locarno"})
```

Przykład 2

- wybierz bazę `employees`

```
use employees;

db;
```

- wstaw dokument do kolekcji `employees`

```
db.employees.insertOne(
    {
        "EmployeeID": 1,
        "FirstName": "Nancy",
        "LastName": "Davolio",
        "Address" : {
            "Street": "507 - 20th Ave. E. Apt. 2A",
            "City": "Seattle",
            "Country": "USA"
        },
        "Title": "Sales Representative",
        "TitleOfCourtesy": "Ms.",
        "BirthDate": ISODate("1948-12-08T00:00:00.000Z"),
        "HireDate": ISODate("1992-05-01T00:00:00.000Z"),
        "Phone": ["(206) 555-9857", "(206) 555-9858"],
        "Salary": 1000
    }
)
```

- sprawdź zawartość kolekcji

```
db.employees.find();
```

- wstaw kilka kolejnych dokumentów za pomocą `insertMany`

```
db.employees.insertMany(
[
    {
        "EmployeeID": 2,
         "FirstName": "Andrew",
        "LastName": "Fuller",
        "Address" : {
            "Street": "908 W. Capital Way",
            "City": "Tacoma",
            "Country": "USA",
        },
        "Title": "Vice President, Sales",
        "TitleOfCourtesy": "Dr.",
        "BirthDate": ISODate("1952-02-19T00:00:00.000Z"),
        "HireDate": ISODate("1992-08-14T00:00:00.000Z"),
        "Phone": ["(206) 555-9482"],
        "Salary": 10000
    },
    {
        "EmployeeID": 3,
        "FirstName": "Janet",
        "LastName": "Leverling",
        "Address" : {
            "Street": "722 Moss Bay Blvd.",
            "City": "Kirkland",
            "Country": "USA",
         },
        "Title": "Sales Representative",
        "TitleOfCourtesy": "Ms.",
        "BirthDate": ISODate("1963-08-30T00:00:00.000Z"),
        "HireDate": ISODate("1992-04-01T00:00:00.000Z"),
        "Phone": ["(206) 555-3412"],
        "Salary": 1200
    },
    {
        "EmployeeID": 4,
        "FirstName": "Margaret",
        "LastName": "Peacock",
        "Address" : {
            "Street": "4110 Old Redmond Rd.",
            "City": "Redmond",
            "Country": "USA",
        },
        "Title": "Sales Representative",
        "TitleOfCourtesy": "Mrs.",
        "BirthDate": ISODate("1937-09-19T00:00:00.000Z"),
        "HireDate": ISODate("1993-05-03T00:00:00.000Z"),
        "Phone": ["(206) 555-8122"],
        "Salary": 1100
    },
     {
        "EmployeeID": 5,
```

```
        "FirstName": "Steven",
        "LastName": "Buchanan",
        "Address" : {
            "Street": "14 Garrett Hill",
            "City": "London",
            "Country": "UK",
        },
        "Title": "Sales Manager",
        "TitleOfCourtesy": "Mr.",
        "BirthDate": ISODate("1955-03-04T00:00:00.000Z"),
        "HireDate": ISODate("1993-10-17T00:00:00.000Z"),
        "Phone": ["(71) 555-4848"],
        "Salary": 2000
    },
    {
        "EmployeeID": 6,
        "FirstName": "Michael",
        "LastName": "Suyama",
        "Address" : {
            "Street": "Coventry House Miner Rd.",
            "City": "London",
            "Country": "UK",
        },
        "Title": "Sales Representative",
        "TitleOfCourtesy": "Mr.",
        "BirthDate": ISODate("1963-07-02T00:00:00.000Z"),
        "HireDate": ISODate("1993-10-17T00:00:00.000Z"),
        "Phone": ["(71) 555-7773"],
        "Salary": 1500
    },
    {
        "EmployeeID": 7,
        "FirstName": "Robert",
        "LastName": "King",
        "Address" : {
            "Street":  "Edgeham Hollow Winchester Way",
            "City": "London",
            "Country": "UK",
        },
        "Title": "Sales Representative",
        "TitleOfCourtesy": "Mr.",
        "BirthDate": ISODate("1960-05-29T00:00:00.000Z"),
        "HireDate": ISODate("1994-01-02T00:00:00.000Z"),
        "Phone": ["(71) 555-5598"],
        "Salary": 1000
    },
    {
        "EmployeeID": 8,
        "FirstName": "Laura",
        "LastName": "Callahan",
        "Address" : {
            "Street":  "4726 - 11th Ave. N.E.",
            "City": "Seattle",
```

```
            "Country": "USA",
        },
        "Title": "Inside Sales Coordinator",
        "TitleOfCourtesy": "Ms.",
        "BirthDate": ISODate("1958-01-09T00:00:00.000Z"),
        "HireDate": ISODate("1994-03-05T00:00:00.000Z"),
        "Phone": ["(206) 555-1189"],
        "Salary": 3000
    },
    {
        "EmployeeID": 9,
        "FirstName": "Anne",
        "LastName": "Dodsworth",
        "Address" : {
            "Street": "7 Houndstooth Rd.",
            "City": "London",
            "Country": "UK",
        },
        "Title": "Sales Representative",
        "TitleOfCourtesy": "Ms.",
        "BirthDate": ISODate("1966-01-27T00:00:00.000Z"),
        "HireDate": ISODate("1994-11-15T00:00:00.000Z"),
        "Phone": ["(71) 555-4444"],
        "Salary": 1400
    }
]
)
```

- w `DataGrip` możesz posłużyć się poleceniami `SQL`
    - podzbiór poleceń `SQL` - proste polecenia `SELECT`
- wykonaj kilka takich poleceń
    - przeanalizuj wyniki
- np.

```
select * from employees;
select * from employees where Address.Country = "USA";
select * from employees where Address.Country = "USA" and TitleOfCourtesy =
"Ms.";

select * from employees where Salary > 1200;
```

- napisz odpowiadające im polecenia `find`
- np

```
db.employees.find({})
db.employees.find({"Address.Country": "USA"})
db.employees.find({$and: [{"Address.Country": "USA"}, {"TitleOfCourtesy" :
"Ms."}]})

db.employees.find(
    {
        $and: [
            {"Address.Country": "USA"},
            {"TitleOfCourtesy" : "Ms."}
        ]
    }
)

db.employees.find({ Salary: { $gt: 1200 } })
```

- operacja projekcji (wybór atrybutów w zbiorze wynikowym)

```
db.employees.find({})
```

```
db.employees.find(
    {},
    {"FirstName": 1, "LastName":1}
)
```

- wynik

```
[
  {
    "_id": {"$oid": "67e879dbb1c69e3d7e42c9c3"},
    "FirstName": "Nancy",
    "LastName": "Davolio"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c5"},
    "FirstName": "Andrew",
    "LastName": "Fuller"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c6"},
    "FirstName": "Janet",
    "LastName": "Leverling"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c7"},
```

```
    "FirstName": "Margaret",
    "LastName": "Peacock"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c8"},
    "FirstName": "Steven",
    "LastName": "Buchanan"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c9"},
    "FirstName": "Michael",
    "LastName": "Suyama"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9ca"},
    "FirstName": "Robert",
    "LastName": "King"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cb"},
    "FirstName": "Laura",
    "LastName": "Callahan"
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cc"},
    "FirstName": "Anne",
    "LastName": "Dodsworth"
  }
]
```

- warunek i projekcja

```
db.employees.find(
    {
        "Address.Country": "USA"
    },
    {
        "FirstName": 1,
        "LastName":1
    }
)
```

- wynik

```
[
  {
    "FirstName": "Nancy",
    "LastName": "Davolio"
```

```
  },
  {
    "FirstName": "Andrew",
    "LastName": "Fuller"
  },
  {
    "FirstName": "Janet",
    "LastName": "Leverling"
  },
  {
    "FirstName": "Margaret",
    "LastName": "Peacock"
  },
  {
    "FirstName": "Laura",
    "LastName": "Callahan"
  }
]
```

- inne przykłady
  - 1 - atrybut pojawi się w zbiorze wynikowym
  - 0 - atrybut nie pojawi się
  - mogą wystąpić albo 1 albo 0
    - nie dotyczy atrybutu `_id_`

```
db.employees.find({},{"_id": 0, "FirstName": 1, "LastName": 1 })

db.employees.find({},{"_id": 1, "FirstName": 0, "LastName": 0})

// to jest błąd
db.employees.find({},{"_id": 1, "FirstName": 1, "LastName": 0})
```

- `count`

```
select count(*) from employees;
select count(*) from employees where Address.Country = "USA";
select count(*) from employees where Address.Country = "USA" and
TitleOfCourtesy = "Ms.";

select count(*) from employees where Salary > 1200;
```

```
db.employees.find({}).count();
db.employees.find({"Address.Country": "USA"}).count();

db.employees.find({$and: [{"Address.Country": "USA"}, {"TitleOfCourtesy" :
```

```
"Ms."}]}).count();


db.employees.find(
    {
        $and: [
            {"Address.Country": "USA"},
            {"TitleOfCourtesy" : "Ms."}
        ]
     }
).count();
```

```
db.employees.find({ Salary: { $gt: 1200 } }).count();
```

- wynik

```
[
  {
    "result": 5
  }
]
```

- $or $and

```
select * from employees where Salary > 1100  and Salary < 1500
```

```
db.employees.find({
  Salary: { $gt: 1100, $lt: 1500 }
})

db.employees.find(
    {
        $and: [
            { Salary: { $gt: 1100 } },
            { Salary: { $lt: 1500 } }
        ]
    }
);
```

- wynik

```
[
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c6"},
    "Address": {
      "Street": "722 Moss Bay Blvd.",
      "City": "Kirkland",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1963-08-30T00:00:00.000Z"},
    "EmployeeID": 3,
    "FirstName": "Janet",
    "HireDate": {"$date": "1992-04-01T00:00:00.000Z"},
    "LastName": "Leverling",
    "Phone": ["(206) 555-3412"],
    "Salary": 1200,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cc"},
    "Address": {
      "Street": "7 Houndstooth Rd.",
      "City": "London",
      "Country": "UK"
    },
    "BirthDate": {"$date": "1966-01-27T00:00:00.000Z"},
    "EmployeeID": 9,
    "FirstName": "Anne",
    "HireDate": {"$date": "1994-11-15T00:00:00.000Z"},
    "LastName": "Dodsworth",
    "Phone": ["(71) 555-4444"],
    "Salary": 1400,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  }
]
```

```
select * from employees where Salary <= 1100  or Salary >= 1500
```

```
db.employees.find(
    {
        $or: [
            { Salary: { $lte: 1100 } },
            { Salary: { $gte: 1500 } }
        ]
    }
);
```

- wynik


```
[
  {
    "_id": {"$oid": "67e879dbb1c69e3d7e42c9c3"},
    "Address": {
      "Street": "507 - 20th Ave. E.\r\nApt. 2A",
      "City": "Seattle",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1948-12-08T00:00:00.000Z"},
    "EmployeeID": 1,
    "FirstName": "Nancy",
    "HireDate": {"$date": "1992-05-01T00:00:00.000Z"},
    "LastName": "Davolio",
    "Phone": ["(206) 555-9857", "(206) 555-9858"],
    "Salary": 1000,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c5"},
    "Address": {
      "Street": "908 W. Capital Way",
      "City": "Tacoma",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1952-02-19T00:00:00.000Z"},
    "EmployeeID": 2,
    "FirstName": "Andrew",
    "HireDate": {"$date": "1992-08-14T00:00:00.000Z"},
    "LastName": "Fuller",
    "Phone": ["(206) 555-9482"],
    "Salary": 10000,
    "Title": "Vice President, Sales",
    "TitleOfCourtesy": "Dr."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c7"},
    "Address": {
      "Street": "4110 Old Redmond Rd.",
      "City": "Redmond",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1937-09-19T00:00:00.000Z"},
    "EmployeeID": 4,
    "FirstName": "Margaret",
    "HireDate": {"$date": "1993-05-03T00:00:00.000Z"},
    "LastName": "Peacock",
    "Phone": ["(206) 555-8122"],
```

```
      "Salary": 1100,
      "Title": "Sales Representative",
      "TitleOfCourtesy": "Mrs."
    },
    {
      "_id": {"$oid": "67e879e2b1c69e3d7e42c9c8"},
      "Address": {
        "Street": "14 Garrett Hill",
        "City": "London",
        "Country": "UK"
      },
      "BirthDate": {"$date": "1955-03-04T00:00:00.000Z"},
      "EmployeeID": 5,
      "FirstName": "Steven",
      "HireDate": {"$date": "1993-10-17T00:00:00.000Z"},
      "LastName": "Buchanan",
      "Phone": ["(71) 555-4848"],
      "Salary": 2000,
      "Title": "Sales Manager",
      "TitleOfCourtesy": "Mr."
    },
    {
      "_id": {"$oid": "67e879e2b1c69e3d7e42c9c9"},
      "Address": {
        "Street": "Coventry House Miner Rd.",
        "City": "London",
        "Country": "UK"
      },
      "BirthDate": {"$date": "1963-07-02T00:00:00.000Z"},
      "EmployeeID": 6,
      "FirstName": "Michael",
      "HireDate": {"$date": "1993-10-17T00:00:00.000Z"},
      "LastName": "Suyama",
      "Phone": ["(71) 555-7773"],
      "Salary": 1500,
      "Title": "Sales Representative",
      "TitleOfCourtesy": "Mr."
    },
    {
      "_id": {"$oid": "67e879e2b1c69e3d7e42c9ca"},
      "Address": {
        "Street": "Edgeham Hollow Winchester Way",
        "City": "London",
        "Country": "UK"
      },
      "BirthDate": {"$date": "1960-05-29T00:00:00.000Z"},
      "EmployeeID": 7,
      "FirstName": "Robert",
      "HireDate": {"$date": "1994-01-02T00:00:00.000Z"},
      "LastName": "King",
      "Phone": ["(71) 555-5598"],
      "Salary": 1000,
      "Title": "Sales Representative",
```

```
    "TitleOfCourtesy": "Mr."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9cb"},
    "Address": {
      "Street": "4726 - 11th Ave. N.E.",
      "City": "Seattle",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1958-01-09T00:00:00.000Z"},
    "EmployeeID": 8,
    "FirstName": "Laura",
    "HireDate": {"$date": "1994-03-05T00:00:00.000Z"},
    "LastName": "Callahan",
    "Phone": ["(206) 555-1189"],
    "Salary": 3000,
    "Title": "Inside Sales Coordinator",
    "TitleOfCourtesy": "Ms."
  }
]
```

```
select * from employees where year(HireDate) = 1992
```

```
db.employees.find(
    {
        $expr: {
            $eq: [ { $year: "$HireDate" }, 1992 ]
        }
    }
);


db.employees.find(
    {
        HireDate: {
            $gte: ISODate("1992-01-01T00:00:00Z"),
            $lt: ISODate("1993-01-01T00:00:00Z")
        }
    }
);
```

- wynik

```
[
  {
    "_id": {"$oid": "67e879dbb1c69e3d7e42c9c3"},
```

```
    "Address": {
      "Street": "507 - 20th Ave. E.\r\nApt. 2A",
      "City": "Seattle",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1948-12-08T00:00:00.000Z"},
    "EmployeeID": 1,
    "FirstName": "Nancy",
    "HireDate": {"$date": "1992-05-01T00:00:00.000Z"},
    "LastName": "Davolio",
    "Phone": ["(206) 555-9857", "(206) 555-9858"],
    "Salary": 1000,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c5"},
    "Address": {
      "Street": "908 W. Capital Way",
      "City": "Tacoma",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1952-02-19T00:00:00.000Z"},
    "EmployeeID": 2,
    "FirstName": "Andrew",
    "HireDate": {"$date": "1992-08-14T00:00:00.000Z"},
    "LastName": "Fuller",
    "Phone": ["(206) 555-9482"],
    "Salary": 10000,
    "Title": "Vice President, Sales",
    "TitleOfCourtesy": "Dr."
  },
  {
    "_id": {"$oid": "67e879e2b1c69e3d7e42c9c6"},
    "Address": {
      "Street": "722 Moss Bay Blvd.",
      "City": "Kirkland",
      "Country": "USA"
    },
    "BirthDate": {"$date": "1963-08-30T00:00:00.000Z"},
    "EmployeeID": 3,
    "FirstName": "Janet",
    "HireDate": {"$date": "1992-04-01T00:00:00.000Z"},
    "LastName": "Leverling",
    "Phone": ["(206) 555-3412"],
    "Salary": 1200,
    "Title": "Sales Representative",
    "TitleOfCourtesy": "Ms."
  }
]
```

Zadanie 2

- Wykonaj kilka własnych eksperymentów

Zadanie 2 - rozwiązanie - Michał Mąka, code review - Michał Kościanek

> Wyniki:
>
> przykłady, kod, zrzuty ekranów, komentarz ...

```
// bazując na danych z przykładów:
db.employees.find()

db.employees.find({
    FirstName: { $regex: /^M/ }
})

select * from employees where FirstName like 'M%';
// employees> db.getCollection("employees")
//    .find({"FirstName": {$regex: /^M/}})

db.employees.find({
    FirstName: { $regex: /^M/ },    // and
    LastName: { $regex: /^S/}
})

db.employees.find({
    $or: [                          // or
        { FirstName: { $regex: /^M/ }},
        {  LastName: { $regex: /^D/}}
    ]
})

select * from employees
where FirstName like 'M%'
OR LastName LIKE 'D%';
// employees> db.getCollection("employees").find({$or:
//    [{"FirstName": {$regex: /^M/}}, {"LastName": {$regex: /^D/}}]})



db.employees.find({
    "Address.City": "Seattle"
})
// dla złożonych wyszukań, gdzie występuje '.' klucz musi być objęty ""

select * from employees
where Address.City is 'Seattle';
// employees> db.getCollection("employees")
//    .find({"Address.City": {$eq: 'Seattle'}})
```

```
// db.collection.find( <query>, <projection>, <options> )
db.employees.find({},{"_id": 0, "FirstName": 1, "LastName": 1 })

db.employees.find({},{"LastName": 1 })
// porównując dwie ostatnie operacje, możemy zauważyć, że
//    _id, będzie wyświetlane by default, natomiast inne pola są ukryte


db.employees.find({
  // porównywanie liczb (lte - lower than or eq), analogicznie gte
    Salary: {$lte: 1500}
}, {FirstName: 1, LastName: 1, Salary: 1}
)


db.employees.find({
    $and:
    [
        {Salary: {$lte: 1500}},
        {Salary: {$gte: 1500}}
        ]
}, {FirstName: 1, LastName: 1, Salary: 1}
)

select FirstName, LastName, Salary
from employees where Salary >= 1500 and Salary <= 1500;
// employees> db.getCollection("employees").find({$and:
//      [{"Salary": {$gte: 1500}}, {"Salary": {$lte: 1500}}]},
//      {"FirstName": 1, "LastName": 1, "Salary": 1, "_id": 0})

// w obydwu przypadkach zgłoszone wykonane polecenie,
//      nie sugeruje automatycznych optymalizacji
//      (check Salary == 1500)
```

Operacje agregacji

- aggregate
- aggregation pipline
- https://www.mongodb.com/docs/manual/core/aggregation-pipeline/

```
db.<collection>.aggregate(

[

   {stage 1},
```

```
    {stage 2},

     …..

    {stage N}

]

)
```

- https://www.mongodb.com/docs/manual/reference/operator/aggregation-pipeline/#std-label-aggregation-pipeline-operator-reference
- Stages
  - `$match`
  - `$project`
  - `$group`
  - `$unwind`
  - `$lookup`
  - `$out`
  - …
- Aggregation vs SQL
  - https://www.mongodb.com/docs/manual/reference/sql-aggregation-comparison/

Przykład 3

- baza `employees` z poprzedniego przykładu

```
use employees;

db;

db.employees.find();
```

- Operacje agregacji
  - `aggregate`

```
db.employees.aggregate([
    {
        $match: {
            "Address.Country": "USA"
        }
```

```
    },
    {
        $project: {
            "_id": 0,
            "LastName": 1,
            "FirstName": 1}
    },
    {
        $sort: {LastName: -1}
    }
])
```

- wynik

```
[
  {
    "FirstName": "Nancy",
    "LastName": "Davolio"
  },
  {
    "FirstName": "Andrew",
    "LastName": "Fuller"
  },
  {
    "FirstName": "Janet",
    "LastName": "Leverling"
  },
  {
    "FirstName": "Margaret",
    "LastName": "Peacock"
  },
  {
    "FirstName": "Laura",
    "LastName": "Callahan"
  }
]
```

- dodatkowe pole `FullName`

```
db.employees.aggregate([
    {
        $match: {
            "Address.Country": "USA"
        }
    },
    {
        $project: {
            "_id": 0,
```

```
            "First": "$LastName",
            "Last": "$LastName",
            "FullName": {
                $concat: ["$FirstName", " ", "$LastName"]
            }

        }
    },
    {
        $sort: {lastname: -1}
    }
])
```

- wynik

```
[
  {
    "First": "Davolio",
    "FullName": "Nancy Davolio",
    "Last": "Davolio"
  },
  {
    "First": "Fuller",
    "FullName": "Andrew Fuller",
    "Last": "Fuller"
  },
  {
    "First": "Leverling",
    "FullName": "Janet Leverling",
    "Last": "Leverling"
  },
  {
    "First": "Peacock",
    "FullName": "Margaret Peacock",
    "Last": "Peacock"
  },
  {
    "First": "Callahan",
    "FullName": "Laura Callahan",
    "Last": "Callahan"
  }
]
```

- grupowanie
    - `$group`

```
select TitleOfCourtesy, count(*)
from employees
```

```
where  Address.Country = "USA"
group by TitleOfCourtesy
```

```
db.employees.aggregate([
    {
        $match: {
            "Address.Country": "USA"
        }
    },
    {
        $group: {
            "_id": "$TitleOfCourtesy",
            "cnt": {"$sum": 1}
        }
    }
])
```

- wynik

```
[
  {
    "_id": "Dr.",
    "cnt": 1
  },
  {
    "_id": "Mrs.",
    "cnt": 1
  },
  {
    "_id": "Ms.",
    "cnt": 3
  }
]
```

- dodatkowa projekcja, zmiana nazwy

```
db.employees.aggregate([
    {
        $match: {
            "Address.Country": "USA"
        }
    },
    {
        $group: {
            "_id": "$TitleOfCourtesy",
            "cnt": {"$sum": 1}
```

```
        }
    },
    {
        $project: {
            "_id": 0,
            "TitleOfCourtesy": "$_id",
            "cnt": 1}
    },

])
```

- grupowanie wg kilku atrybutów

```
select Address.Country, TitleOfCourtesy, count(*) cnt
from employees
group by Address.Country, TitleOfCourtesy
order by cnt desc;
```

```
db.employees.aggregate([
    {
        $group: {
            _id: {
                country: "$Address.Country",
                title: "$TitleOfCourtesy"
            },
            cnt: { $sum: 1 }
        }
    },
    {
        $project: {
        _id: 0,
        country: "$_id.country",
        title: "$_id.title",
        cnt: 1
        }
    },
    {
        $sort: {cnt: -1}
    }
])
```

- wynik

```
[
  {
    "Country": "UK",
```

```
    "TitleOfCourtesy": "Mr.",
    "cnt": 3
  },
  {
    "Country": "USA",
    "TitleOfCourtesy": "Ms.",
    "cnt": 3
  },
  {
    "Country": "USA",
    "TitleOfCourtesy": "Dr.",
    "cnt": 1
  },
  {
    "Country": "UK",
    "TitleOfCourtesy": "Ms.",
    "cnt": 1
  },
  {
    "Country": "USA",
    "TitleOfCourtesy": "Mrs.",
    "cnt": 1
  }
]
```

- zapis wyniku
  - `$out`
  - zapis zbioru wynikowego to kolekcji o wskazanej nazwie

```
db.employees.aggregate([
    {
        $group: {
            _id: {
                Country: "$Address.Country",
                TitleOfCourtesy: "$TitleOfCourtesy"
            },
            cnt: { $sum: 1 }
        }
    },
    {
        $project: {
        _id: 0,
        Country: "$_id.Country",
        TitleOfCourtesy: "$_id.TitleOfCourtesy",
        cnt: 1
        }
    },
    {
        $out: "em_by_country_title"
```

```
    }
])
```

- lista kolekcji

```
show collections;
```

- wynik

```
[
  {
    "badge": "",
    "name": "em_by_country_title"
  },
  {
    "badge": "",
    "name": "employees"
  }
]
```

- zawartość kolekcji `em_by_country_title`

```
db.em_by_country_title.find();
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec20f77b1ef941c9512dcb"},
    "Country": "USA",
    "TitleOfCourtesy": "Mrs.",
    "cnt": 1
  },
  {
    "_id": {"$oid": "67ec20f77b1ef941c9512dcc"},
    "Country": "USA",
    "TitleOfCourtesy": "Ms.",
    "cnt": 3
  },
  {
    "_id": {"$oid": "67ec20f77b1ef941c9512dcd"},
    "Country": "USA",
    "TitleOfCourtesy": "Dr.",
    "cnt": 1
```

```
  },
  {
    "_id": {"$oid": "67ec20f77b1ef941c9512dce"},
    "Country": "UK",
    "TitleOfCourtesy": "Mr.",
    "cnt": 3
  },
  {
    "_id": {"$oid": "67ec20f77b1ef941c9512dcf"},
    "Country": "UK",
    "TitleOfCourtesy": "Ms.",
    "cnt": 1
  }
]
```

Zadanie 3

- Wykonaj kilka własnych eksperymentów

Zadanie 3 - rozwiązanie - Michał Kościanek, code review - Michał Mąka

> Wyniki:
>
> przykłady, kod, zrzuty ekranów, komentarz ...

```
//wynik to posortowana malejąco liczba pracowników z każdego miasta
db.employees.aggregate([
    {
      //group by jak z T-SQL,
      //
        $group: {
            "_id": "$City",
            // do każdego zgrupowanego dokumentu dodaj 1
            "LiczbaPracownikow": { $sum: 1 }
        }
    },
    {
      //sortowanie DESC
        $sort: { "LiczbaPracownikow": -1 }
    }
])
//wynik to reprezentacja ilu pracowników i jacy pracownicy pracują
// z danego państwa
db.employees.aggregate([
    {
      //grupujemy po krajach
        $group: {
            "_id": "$Country",
```

```
            //dodajemy do wyniku tabele pasujących pracowników
            "Pracownicy": { $push: "$LastName" },
            //podliczamy każdego dodanego
            "Liczba": { $sum: 1 }
        }
    },
    {
      //zaprojektowane wyświetlanie się wyników
        $project: {
            "_id": 0,
            "Kraj": "$_id",
            "Pracownicy": 1,
            "Liczba": 1
        }
    }
])
// wynik to 2 pierwszych wg. alfabetu po nazwisku pracownicy
// pomijając 3 pierwsze wyniki
db.employees.aggregate([
    {
        $project: {
            "_id": 0,
            "Imie": "$FirstName",
            "Nazwisko": "$LastName"
        }
    },
    {
        $sort: { "Nazwisko": 1 }
    },
    {
        $skip: 3
    },
    {
        $limit: 2
    }
])
```

Przykład 4

Odtwórz z backupu bazę `north0`

- plik z z backup'em bazy danych dostępny jest w moodle
- najprostsza wersja
    - `mongorestore <folder>`

```
mongorestore dump
```

- to polecenie odtworzy wszystkie bazy danych znajdujące się we wskazanym folderze (w tym przypadku `dump`)
    - najłatwiej wgrać tam folder zawierający pliki z backupem i wykonać proste polecenie mongorestore
- dokumentacja:
    - https://www.mongodb.com/docs/database-tools/mongorestore/

Wybierz bazę north0

```
use north0
```

Baza `north0` jest kopią relacyjnej bazy danych `Northwind`

- poszczególne kolekcje odpowiadają tabelom w oryginalnej bazie `Northwind`

zapoznaj się ze strukturą dokumentów w bazie `North0`

```
db.customers.find()
db.orders.find();
db.orderdetails.find();
```

Przyklad 5

produkty, kategorie, dostawcy

```
db.categories.find().limit(2);
```

- wynik

```
[
  {
    "_id": {"$oid": "67eb06a27b1ef941c950fb49"},
    "CategoryID": 1,
    "CategoryName": "Beverages",
    "Description": "Soft drinks, coffees, teas, beers, and ales"
  },
  {
    "_id": {"$oid": "67eb06a27b1ef941c950fb4a"},
    "CategoryID": 2,
    "CategoryName": "Condiments",
```

```
    "Description": "Sweet and savory sauces, relishes, spreads, and seasonings"
  }
]
```

```
db.products.find().limit(2);
```

- wynik

```
[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "SupplierID": 1,
    "UnitPrice": 18,
    "UnitsInStock": 39,
    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
    "SupplierID": 1,
    "UnitPrice": 19,
    "UnitsInStock": 17,
    "UnitsOnOrder": 40
  }
]
```

```
db.suppliers.find().limit(2);
```

- wynik

```
[
  {
    "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
```

```
      "Address": "49 Gilbert St.",
      "City": "London",
      "CompanyName": "Exotic Liquids",
      "ContactName": "Charlotte Cooper",
      "ContactTitle": "Purchasing Manager",
      "Country": "UK",
      "Fax": null,
      "HomePage": null,
      "Phone": "(171) 555-2222",
      "PostalCode": "EC1 4SD",
      "Region": null,
      "SupplierID": 1
    },
    {
      "_id": {"$oid": "63a05d3cbb3b972d6f4e09f8"},
      "Address": "P.O. Box 78934",
      "City": "New Orleans",
      "CompanyName": "New Orleans Cajun Delights",
      "ContactName": "Shelley Burke",
      "ContactTitle": "Order Administrator",
      "Country": "USA",
      "Fax": null,
      "HomePage": "#CAJUN.HTM#",
      "Phone": "(100) 555-4822",
      "PostalCode": "70117",
      "Region": "LA",
      "SupplierID": 2
    }
  ]
```

Uwaga:

- etap `$match` z pustym warunkiem zwraca wszystkie dokumenty (nie jest potrzebny, można go pominąć)
    - w przykładzie etap został dodany żeby łatwiej można było wykonać własne eksperymenty ograniczające zbiór dokumentów z kolekcji `products`

```
db.products.aggregate([
    {
        $match : { }
    },

    //suppliers
    {
        $lookup : {
            from: "suppliers",
            localField: "SupplierID",
            foreignField: "SupplierID",
            as: "Supplier"
        }
```

```
    },
    {
        $limit : 2
    }
])
```

- pole `Suplier` jest tablicą
    - zawiera 1 obiekt
- wynik


```
[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": [
      {
        "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
        "SupplierID": 1,
        "CompanyName": "Exotic Liquids",
        "ContactName": "Charlotte Cooper",
        "ContactTitle": "Purchasing Manager",
        "Address": "49 Gilbert St.",
        "City": "London",
        "Region": null,
        "PostalCode": "EC1 4SD",
        "Country": "UK",
        "Phone": "(171) 555-2222",
        "Fax": null,
        "HomePage": null
      }
    ],
    "SupplierID": 1,
    "UnitPrice": 18,
    "UnitsInStock": 39,
    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
```

```
    "Supplier": [
      {
        "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
        "SupplierID": 1,
        "CompanyName": "Exotic Liquids",
        "ContactName": "Charlotte Cooper",
        "ContactTitle": "Purchasing Manager",
        "Address": "49 Gilbert St.",
        "City": "London",
        "Region": null,
        "PostalCode": "EC1 4SD",
        "Country": "UK",
        "Phone": "(171) 555-2222",
        "Fax": null,
        "HomePage": null
      }
    ],
    "SupplierID": 1,
    "UnitPrice": 19,
    "UnitsInStock": 17,
    "UnitsOnOrder": 40
  }
]
```

- `$unwind`
  - pole `Supplier` jest obiektem/dokumentem

```
db.products.aggregate([
    {
        $match : { }
    },

    //suppliers
    {
        $lookup : {
            from: "suppliers",
            localField: "SupplierID",
            foreignField: "SupplierID",
            as: "Supplier"
        }
    },
    {
        $unwind : "$Supplier"
    },
   {
        $limit : 2
   }
])
```

- wynik


```
[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": {
      "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "ContactName": "Charlotte Cooper",
      "ContactTitle": "Purchasing Manager",
      "Address": "49 Gilbert St.",
      "City": "London",
      "Region": null,
      "PostalCode": "EC1 4SD",
      "Country": "UK",
      "Phone": "(171) 555-2222",
      "Fax": null,
      "HomePage": null
    },
    "SupplierID": 1,
    "UnitPrice": 18,
    "UnitsInStock": 39,
    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
    "Supplier": {
      "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "ContactName": "Charlotte Cooper",
      "ContactTitle": "Purchasing Manager",
      "Address": "49 Gilbert St.",
      "City": "London",
      "Region": null,
      "PostalCode": "EC1 4SD",
      "Country": "UK",
      "Phone": "(171) 555-2222",
      "Fax": null,
```

```
      "HomePage": null
    },
    "SupplierID": 1,
    "UnitPrice": 19,
    "UnitsInStock": 17,
    "UnitsOnOrder": 40
  }
]
```

- projekcja
  - tylko wybrane pola z `Supplier`

```
db.products.aggregate([
    {
        $match : { }
    },

    //suppliers
    {
        $lookup : {
            from: "suppliers",
            localField: "SupplierID",
            foreignField: "SupplierID",
            as: "Supplier_tmp"
        }
    },
    {
        $unwind : "$Supplier_tmp"
    },
    {
        $addFields: {
            Supplier: {
                SupplierID: "$SupplierID",
                CompanyName: "$Supplier_tmp.CompanyName",
                Country: "$Supplier_tmp.Country"
            }
       }
    },
    {
        $project: {
            SupplierID: 0,
            Supplier_tmp: 0
        }
    },
    {
        $limit : 2
    }
])
```

- wynik

```
[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": {
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "Country": "UK"
    },
    "UnitPrice": 18,
    "UnitsInStock": 39,
    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "CategoryID": 1,
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
    "Supplier": {
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "Country": "UK"
    },
    "UnitPrice": 19,
    "UnitsInStock": 17,
    "UnitsOnOrder": 40
  }
]
```

- dodaj do wyniku inf. o kategoriach

```
db.products.aggregate([
    {
        $match : { }
    },

    //suppliers
    {
```

```
        $lookup : {
            from: "suppliers",
            localField: "SupplierID",
            foreignField: "SupplierID",
            as: "Supplier_tmp"
        }
    },
    //categories
    {
        $lookup : {
            from: "categories",
            localField: "CategoryID",
            foreignField: "CategoryID",
            as: "Category_tmp"
        }
    },
    {
        $unwind : "$Supplier_tmp"
    },
    {
        $unwind : "$Category_tmp"
    },
    {
        $addFields: {
            Supplier: {
                SupplierID: "$SupplierID",
                CompanyName: "$Supplier_tmp.CompanyName",
                Country: "$Supplier_tmp.Country"
            },
            Category: {
                CategoryID: "$CategoryID",
                CategorynameName: "$Category_tmp.CategoryName",
            }
        }
    },
    {
        $project: {
            SupplierID: 0,
            Supplier_tmp: 0,
            CategoryID: 0,
            Category_tmp: 0
        }
    },
    {
        $limit : 2
    }
])
```

- wynik

```
[
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f50"},
    "Category": {
      "CategoryID": 1,
      "CategorynameName": "Beverages"
    },
    "Discontinued": false,
    "ProductID": 1,
    "ProductName": "Chai",
    "QuantityPerUnit": "10 boxes x 20 bags",
    "ReorderLevel": 10,
    "Supplier": {
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "Country": "UK"
    },
    "UnitPrice": 18,
    "UnitsInStock": 39,
    "UnitsOnOrder": 0
  },
  {
    "_id": {"$oid": "63a0607abb3b972d6f4e1f51"},
    "Category": {
      "CategoryID": 1,
      "CategorynameName": "Beverages"
    },
    "Discontinued": false,
    "ProductID": 2,
    "ProductName": "Chang",
    "QuantityPerUnit": "24 - 12 oz bottles",
    "ReorderLevel": 25,
    "Supplier": {
      "SupplierID": 1,
      "CompanyName": "Exotic Liquids",
      "Country": "UK"
    },
    "UnitPrice": 19,
    "UnitsInStock": 17,
    "UnitsOnOrder": 40
  }
]
```


Przykład 6

- dostawcy i produkty

```
db.suppliers.aggregate([
    {
        $match : { }
    },

    //products
    {
        $lookup : {
            from: "products",
            localField: "SupplierID",
            foreignField: "SupplierID",
            as: "Products"
        }
    },
    {
        $addFields: {
            Products: {
                $map: {
                    input: "$Products",
                    as: "product",
                    in: {
                        ProductID: "$$product.ProductID",
                        ProductName: "$$product.ProductName",
                        UnitPrice: "$$product.UnitPrice",
                        Discontinued: "$$product.Discontinued",
                    }
                }
            }
        }
    },
    {
        $limit : 2
    }
])
```

- wynik

```
[
  {
    "_id": {"$oid": "63a05d3cbb3b972d6f4e09f7"},
    "Address": "49 Gilbert St.",
    "City": "London",
    "CompanyName": "Exotic Liquids",
    "ContactName": "Charlotte Cooper",
    "ContactTitle": "Purchasing Manager",
    "Country": "UK",
    "Fax": null,
    "HomePage": null,
    "Phone": "(171) 555-2222",
    "PostalCode": "EC1 4SD",
```

```
      "Products": [
        {
          "ProductID": 1,
          "ProductName": "Chai",
          "UnitPrice": 18,
          "Discontinued": false
        },
        {
          "ProductID": 2,
          "ProductName": "Chang",
          "UnitPrice": 19,
          "Discontinued": false
        },
        {
          "ProductID": 3,
          "ProductName": "Aniseed Syrup",
          "UnitPrice": 10,
          "Discontinued": false
        }
      ],
      "Region": null,
      "SupplierID": 1
    },
    {
      "_id": {"$oid": "63a05d3cbb3b972d6f4e09f8"},
      "Address": "P.O. Box 78934",
      "City": "New Orleans",
      "CompanyName": "New Orleans Cajun Delights",
      "ContactName": "Shelley Burke",
      "ContactTitle": "Order Administrator",
      "Country": "USA",
      "Fax": null,
      "HomePage": "#CAJUN.HTM#",
      "Phone": "(100) 555-4822",
      "PostalCode": "70117",
      "Products": [
        {
          "ProductID": 4,
          "ProductName": "Chef Anton's Cajun Seasoning",
          "UnitPrice": 22,
          "Discontinued": false
        },
        {
          "ProductID": 5,
          "ProductName": "Chef Anton's Gumbo Mix",
          "UnitPrice": 21.35,
          "Discontinued": true
        },
        {
          "ProductID": 65,
          "ProductName": "Louisiana Fiery Hot Pepper Sauce",
          "UnitPrice": 21.05,
          "Discontinued": false
```

```
      },
      {
        "ProductID": 66,
        "ProductName": "Louisiana Hot Spiced Okra",
        "UnitPrice": 17,
        "Discontinued": false
      }
    ],
    "Region": "LA",
    "SupplierID": 2
  }
]
```

Przyklad 7

- produkty
  - tworzymy kolekcję `products_tmp` (połączenie danych z klolekcji `products, suppliers, categories`)

```
db.products.aggregate([

   { $match : { } }

   //suppliers
   , {$lookup : {
                 from: "suppliers",
                 localField: "SupplierID",
                 foreignField: "SupplierID",
                 as: "supplier"
                }

   }
   , {$unwind : "$supplier"}
   , {$project : {
                    "supplier._id": 0
                    ,"supplier.ContactName": 0
                    ,"supplier.ContactTitle": 0
                    ,"supplier.City": 0
                    ,"supplier.Country": 0
                    ,"supplier.Address": 0
                    ,"supplier.PostalCode": 0
                    ,"supplier.Region": 0
                    ,"supplier.Phone": 0
                    ,"supplier.Fax": 0
                    ,"supplier.HomePage": 0
                 }
    }
```

```
     //categories
    , {$lookup : {
                 from: "categories",
                 localField: "CategoryID",
                 foreignField: "CategoryID",
                 as: "category"
                }
    }
    , {$unwind : "$category"}
    , {$project : {
                    "category._id": 0
                    ,"category.Description": 0
                    ,"category.Picture": 0

                 }
     }

    //usunięcie kluczy obcych
   , {$project : {
                    "Discontinued": 0
                    ,"QuantityPerUnit": 0
                    ,"ReorderLevel": 0
                    ,"UnitPrice": 0
                    ,"UnitsInStock": 0
                    ,"UnitsOnOrder": 0
                 }
    }
 , {$project : {
                    "ProductID": 1
                    ,"ProductName": 1
                    ,"CategoryID": 1
                    ,"CategoryName": "$category.CategoryName"
                    ,"SupplierID": 1
                    ,"SupplierName": "$supplier.CompanyName"
                 }
    }

    , {$out : "products_tmp"}
 ]

 )
```

```
db.products_tmp.find();
```

- pozycje zamówień
  - tworzymy kolekcję `orderdetails_tmp` (polecenie wykorzystuje `products_tmp` stworzoną w poprzednim przykładzie)

```
db.orderdetails.aggregate(

    { $match : { } }

   //suppliers
   , {$lookup : {
                 from: "products_tmp",
                 localField: "ProductID",
                 foreignField: "ProductID",
                 as: "product"
                }

   }
   , {$unwind : "$product"}
   , {$project : {
                    "product._id": 0
                 }
    }
    , {$project : {
                    "ProductID": 0
                 }
    }
    , {$addFields : {
                    "TotalValue": {$multiply : ["$UnitPrice", "$Quantity",
{$subtract: [1, "$Discount"]}]}
                    ,"Discount": {$round: ["$Discount",2]}
                 }
    }


, {$out : "orderdetails_tmp"}

)
```

```
db.orderdetails_tmp.find()
```

- zamówienia
  - każdy dokument agreguje pełną informację o zamówieniu
    - tworzymy kolekcję `orders_tmp` (polecenie wykorzystuje `orderdetails_tmp` stworzoną w poprzednim przykładzie)

```
db.orders.aggregate(
    {
        $match : { }
    },

   //customers
```

```
    {
        $lookup : {
            from: "customers",
            localField: "CustomerID",
            foreignField: "CustomerID",
            as: "Customer"
        }

    },
    {
        $unwind : "$Customer"
    },
    {
        $project : {
            "Customer._id": 0,
            "Customer.ContactName": 0,
            "Customer.ContactTitle": 0,
            "Customer.City": 0,
//              "Customer.Country": 0,
            "Customer.Address": 0,
            "Customer.PostalCode": 0,
            "Customer.Region": 0,
            "Customer.Phone": 0,
            "Customer.Fax": 0
        }
    },

    //employees
    {
        $lookup : {
            from: "employees",
                localField: "EmployeeID",
                foreignField: "EmployeeID",
                as: "Employee"
        }
    },
    {
        $unwind : "$Employee"
    },
    {
        $project : {
            "Employee._id": 0,
            "Employee.Title": 0,
            "Employee.TitleOfCourtesy": 0,
            "Employee.BirthDate": 0,
            "Employee.HireDate": 0,
            "Employee.Address": 0,
            "Employee.PostalCode": 0,
            "Employee.City": 0,
            "Employee.Region": 0,
            "Employee.Country": 0,
            "Employee.HomePhone": 0,
            "Employee.Extension": 0,
```

```
            "Employee.Photo": 0,
            "Employee.Notes": 0,
            "Employee.ReportsTo": 0,
            "Employee.PhotoPath": 0
        }
    },
    //shippers
    {
        $lookup : {
            from: "shippers",
            localField: "ShipVia",
            foreignField: "ShipperID",
            as: "Shipper"
        }
    },
    {
        $unwind : "$Shipper"
    },
    {
        $project : {
            "Shipper._id": 0,
            "Shipper.Phone": 0

        }
    },
    //orderdetails
    {
        $lookup : {
            from: "orderdetails_tmp",
            localField: "OrderID",
            foreignField: "OrderID",
            as: "Orderdetails"
        }
    },
    {
        $project : {
              "Orderdetails._id": 0,
              "Orderdetails.OrderID": 0

        }
    },
    {
        $project : {
              "CustomerID": 0,
              "EmployeeID": 0,
              "ShipVia": 0
        }
   },
   {
    $out : "orders_tmp"
   }

)
```

```
db.orders_tmp.find().limit(2);
```

- wynik


```
[
  {
    "_id": {"$oid": "63a060b9bb3b972d6f4e1fc6"},
    "Customer": {
      "CustomerID": "VINET",
      "CompanyName": "Vins et alcools Chevalier",
      "Country": "France"
    },
    "Employee": {
      "EmployeeID": 5,
      "LastName": "Buchanan",
      "FirstName": "Steven"
    },
    "Freight": 32.38,
    "OrderDate": {"$date": "1996-07-04T00:00:00.000Z"},
    "OrderID": 10248,
    "Orderdetails": [
      {
        "UnitPrice": 14,
        "Quantity": 12,
        "Discount": 0,
        "product": {
          "ProductID": 11,
          "ProductName": "Queso Cabrales",
          "QuantityPerUnit": "1 kg pkg."
        },
        "TotalValue": 168
      },
      {
        "UnitPrice": 9.8,
        "Quantity": 10,
        "Discount": 0,
        "product": {
          "ProductID": 42,
          "ProductName": "Singaporean Hokkien Fried Mee",
          "QuantityPerUnit": "32 - 1 kg pkgs."
        },
        "TotalValue": 98
      },
      {
        "UnitPrice": 34.8,
        "Quantity": 5,
        "Discount": 0,
```

```
          "product": {
            "ProductID": 72,
            "ProductName": "Mozzarella di Giovanni",
            "QuantityPerUnit": "24 - 200 g pkgs."
          },
          "TotalValue": 174
        }
      ],
      "RequiredDate": {"$date": "1996-08-01T00:00:00.000Z"},
      "ShipAddress": "59 rue de l'Abbaye",
      "ShipCity": "Reims",
      "ShipCountry": "France",
      "ShipName": "Vins et alcools Chevalier",
      "ShipPostalCode": "51100",
      "ShipRegion": null,
      "ShippedDate": {"$date": "1996-07-16T00:00:00.000Z"},
      "Shipper": {
        "ShipperID": 3,
        "CompanyName": "Federal Shipping"
      }
    },
    {
      "_id": {"$oid": "63a060b9bb3b972d6f4e1fc7"},
      "Customer": {
        "CustomerID": "TOMSP",
        "CompanyName": "Toms Spezialitäten",
        "Country": "Germany"
      },
      "Employee": {
        "EmployeeID": 6,
        "LastName": "Suyama",
        "FirstName": "Michael"
      },
      "Freight": 11.61,
      "OrderDate": {"$date": "1996-07-05T00:00:00.000Z"},
      "OrderID": 10249,
      "Orderdetails": [
        {
          "UnitPrice": 18.6,
          "Quantity": 9,
          "Discount": 0,
          "product": {
            "ProductID": 14,
            "ProductName": "Tofu",
            "QuantityPerUnit": "40 - 100 g pkgs."
          },
          "TotalValue": 167.4
        },
        {
          "UnitPrice": 42.4,
          "Quantity": 40,
          "Discount": 0,
          "product": {
```

```
          "ProductID": 51,
          "ProductName": "Manjimup Dried Apples",
          "QuantityPerUnit": "50 - 300 g pkgs."
        },
        "TotalValue": 1696
      }
    ],
    "RequiredDate": {"$date": "1996-08-16T00:00:00.000Z"},
    "ShipAddress": "Luisenstr. 48",
    "ShipCity": "Münster",
    "ShipCountry": "Germany",
    "ShipName": "Toms Spezialitäten",
    "ShipPostalCode": "44087",
    "ShipRegion": null,
    "ShippedDate": {"$date": "1996-07-10T00:00:00.000Z"},
    "Shipper": {
      "ShipperID": 1,
      "CompanyName": "Speedy Express"
    }
  }
]
```

Zadanie 4

- zmień (popraw) strukturę dokumentu w kolekcji `orders_tmp`
- stwórz obiekt/dokument grupujący inf. o adresie wysyłki
    - np.

```
Shippment : {
    "ShipAddress": "Luisenstr. 48",
    "ShipCity": "Münster",
    "ShipCountry": "Germany",
    "ShipName": "Toms Spezialitäten",
    "ShipPostalCode": "44087",
    "ShipRegion": null,
}
```

- stwórz obiekt/dokument grupujący inf. o datach (zamówienia, wysyłki ...)
    - np.

```
Dates : {
    "OrderDate": {"$date": "1996-07-04T00:00:00.000Z
    "RequiredDate": {"$date": "1996-08-01T00:00:00.000Z"},
    "ShippedDate": {"$date": "1996-07-16T00:00:00.000Z"},
}
```

- Dodaj inf. o pełnej wartości całego zamówienia

```
TotalOrderValue : ...
```

Zadanie 4 - rozwiązanie - Michał Mąka, code review - Michał Kościanek

> Wyniki:
>
> przykłady, kod, zrzuty ekranów, komentarz ...

```
db.orders_tmp.aggregate([
    {
        // uporządkowanie
        $project: {
            OrderID: 1,
            CustomerID: 1,
            EmployeeID: 1,
//            OrderDate: 1,
//            RequiredDate: 1,
//            ShippedDate: 1,
            // powtorzone w dates
            ShipVia: 1,
            Freight: 1,

            Shippment: {
                ShipAddress: "$ShipAddress",
                ShipCity: "$ShipCity",
                ShipCountry: "$ShipCountry",
                ShipName: "$ShipName",
                ShipPostalCode: "$ShipPostalCode",
                ShipRegion: "$ShipRegion"
            },

            Dates: {
                OrderDate: "$OrderDate",
                RequiredDate: "$RequiredDate",
                ShippedDate: "$ShippedDate",
            },
        },
    },
    // aby znaleźć TotalOrderValue musimy znaleźć cene danych produktów
    {
        $lookup: {
            // w orderdetails_tmp mamy juz zsumowane totalValue
            //  dla pojedyńczych produktów w zamówieniu
            from: "orderdetails_tmp",
            localField: "OrderID",
```

```
            foreignField: "OrderID",
            as: "OrderItems"
        }
    },
    {
        $addFields: {
            TotalOrderValue: { $sum: "$OrderItems.TotalValue" }
        }
    },
    // dodatkowo możemy schować zamówienia na pojedyńcze produkty
    {
        $project: {
            OrderItems: 0
        }
    },

    { $out: "orders_tmp2" }
])

db.orderdetails_tmp.find()
db.orders_tmp2.find().limit(1)
```

- Wyniki

```
// bez zschowanych produktów,
//    żeby sprawdzić sumowanie (OrderItems: 1)

[
  {
    "_id": {"$oid": "63a060b9bb3b972d6f4e1fc6"},
    "Dates": {
      "OrderDate": {"$date": "1996-07-04T00:00:00.000Z"},
      "RequiredDate": {"$date": "1996-08-01T00:00:00.000Z"},
      "ShippedDate": {"$date": "1996-07-16T00:00:00.000Z"}
    },
    "Freight": 32.38,
    "OrderID": 10248,
    "OrderItems": [
      {
        "_id": {"$oid": "63a06016bb3b972d6f4e16bc"},
        "OrderID": 10248,
        "UnitPrice": 14,
        "Quantity": 12,
        "Discount": 0,
        "product": {
          "ProductID": 11,
          "ProductName": "Queso Cabrales",
          "SupplierID": 5,
          "CategoryID": 4,
          "CategoryName": "Dairy Products",
          "SupplierName": "Cooperativa de Quesos 'Las Cabras'"
```

```
          },
          "TotalValue": 168
        },
        {
          "_id": {"$oid": "63a06016bb3b972d6f4e16bd"},
          "OrderID": 10248,
          "UnitPrice": 9.8,
          "Quantity": 10,
          "Discount": 0,
          "product": {
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            "ProductName": "Singaporean Hokkien Fried Mee",
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            "CategoryID": 5,
            "CategoryName": "Grains/Cereals",
            "SupplierName": "Leka Trading"
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          "Quantity": 5,
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            "ProductName": "Mozzarella di Giovanni",
            "SupplierID": 14,
            "CategoryID": 4,
            "CategoryName": "Dairy Products",
            "SupplierName": "Formaggi Fortini s.r.l."
          },
          "TotalValue": 174
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        "ShipCity": "Reims",
        "ShipCountry": "France",
        "ShipName": "Vins et alcools Chevalier",
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        "ShipName": "Vins et alcools Chevalier",
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